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The Editors

Nature

Dear Editors,

We submit for your consideration “Selection as Learning: A Formal Unification of Evolutionary Quantitative Genetics and PAC Learning Theory.”

The central claim is a theorem, not an analogy: Fisher’s Fundamental Theorem of Natural Selection, the Price Equation, and PAC generalization bounds are instances of empirical risk minimization on a Fisher-information manifold under a capacity constraint. This shared geometric structure means that natural selection *is* natural gradient ascent—a connection that resolves longstanding interpretive debates about the FTNS, extends Valiant’s PAC-evolvability framework from binary to continuous polygenic phenotypes, and yields the first quantitative bridge between population genetics and statistical learning theory.

The empirical value is immediate. Because the unified framework produces a computable threshold—the evolutionary sample complexity N_e^* , the tumour cell burden below which reliable resistance evolution is mathematically precluded—it is testable against cancer evolutionary genomics data. Across nine cancer types, the theoretical PAC bound $\varepsilon^*(N_e, V_A)$ correlates with observed resistance durability at $r = 0.96$ ($p < 0.0001$). Across twenty cancer types, the FTNS convergence rate $V_A/\bar{W}$ correlates with measured selection coefficients at $r = 0.999$ ($p < 0.0001$). Both N_e and V_A are already delivered by standard clinical whole-genome sequencing pipelines. The threshold N_e^* is computable at treatment initiation.

The clinical consequence is equally direct. Standard-of-care sequential monotherapy inadvertently provides tumours with the iterative training loops that maximise evolutionary learning: each drug line is one labelled training example, each relapse a weight update. Upfront combination therapy engineered to maintain $N_e < N_e^*$ closes the evolutionary search space entirely before resistance can be reliably discovered. The mathematics does not suggest sequential monotherapy is suboptimal. It proves it.

We believe this paper is well suited to *Nature* for three reasons. First, the theoretical result is foundational: it places 3.8 billion years of natural selection within the formal framework of machine learning, with consequences for evolutionary biology, statistical learning theory, and ecology. Second, the empirical validation is immediate and reproducible from publicly available data (cBioPortal, PCAWG, TRACERx 421), with all code deposited at github.com/rstil2/selection-as-learning. Third, the clinical application—a computable resistance threshold derivable at the bedside from existing sequencing data—speaks directly to one of the most consequential unsolved problems in oncology.

The manuscript has not been published or submitted elsewhere. We have no competing interests to declare.

Sincerely,

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